# Hierarchical EMFI Analysis on a RISC-V SoC

*From a black-box die to a PDN-localised first bit-flip – 12 756 flip-flops, 0.18 µm CMOS*


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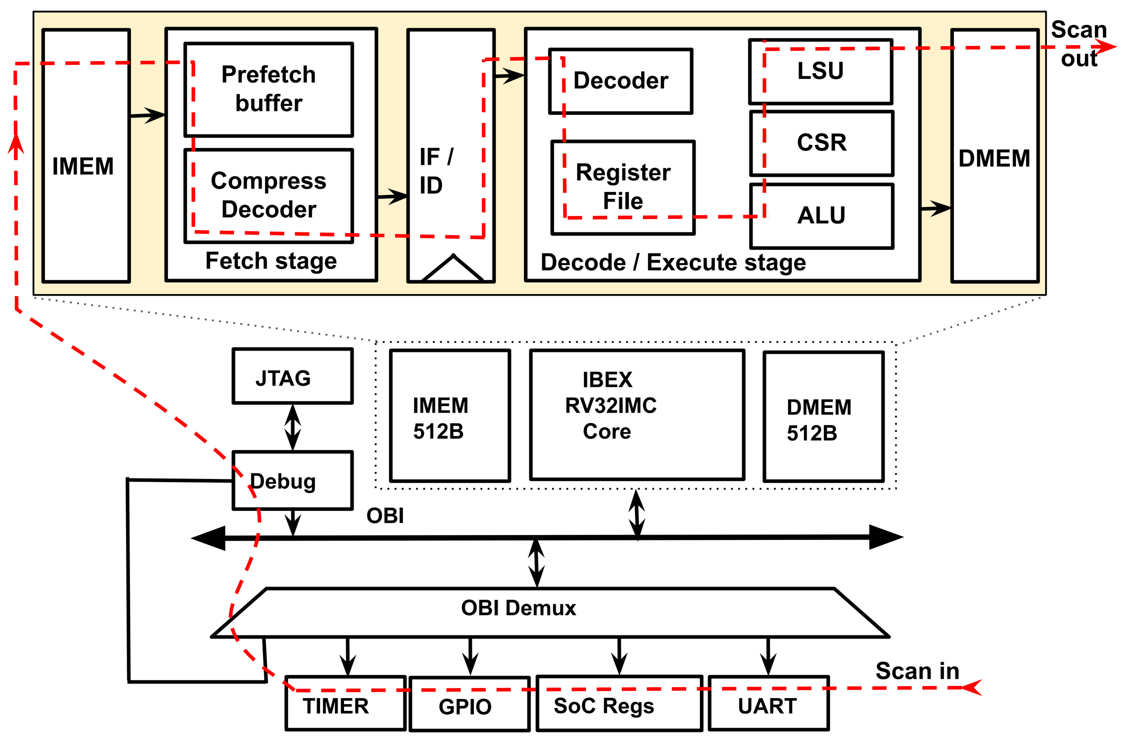
## 1. Problem: The die is a black box

- EMFI is a practical attack vector, but is hard to analyze at the design level.
- EMFI flips bits *inside* the die; bench tester sees only externals.
- Simulation assumes fault models; misses multi-bit upsets & PDN coupling.

**Observability gap = root-cause gap.**

## 2. Solution: scan chain as observability



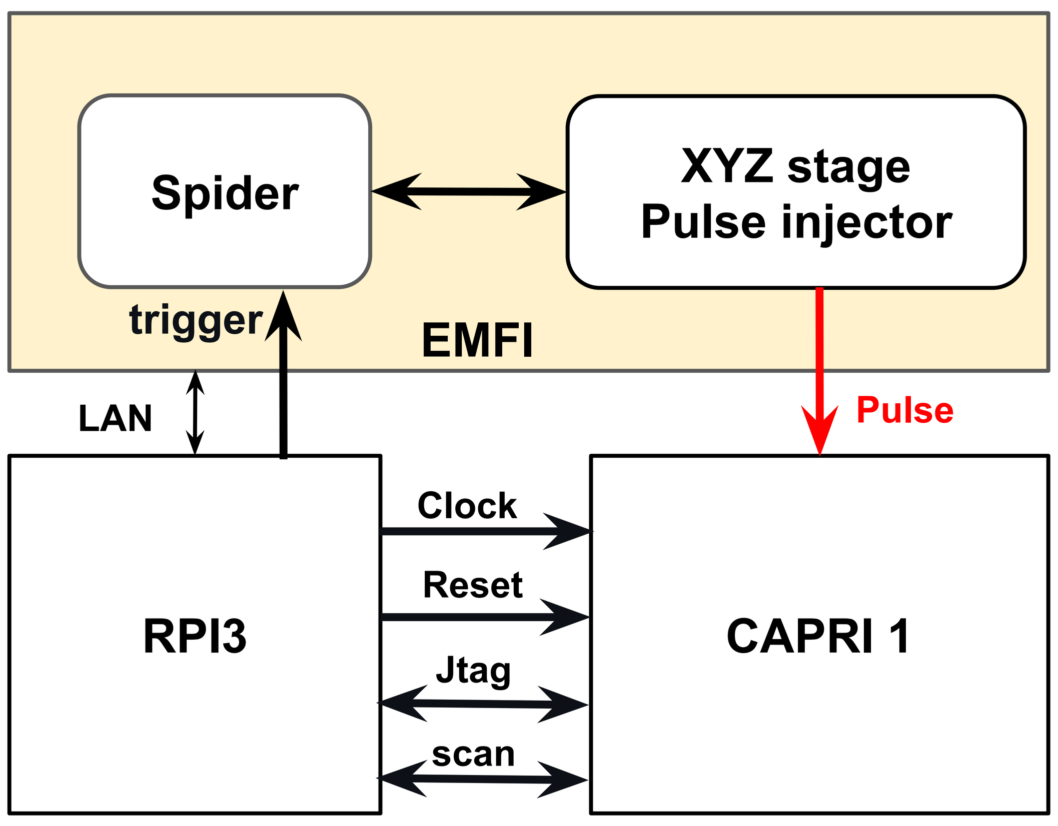

Red dashed = scan path through every FF.

**All 12 756 FFs** dumped per shot: halt → shift out → diff vs. golden run.

**One scan dump ⇒ three answers:**

- **Where** – first-flip cell, $(x, y)$.
- **How** – µarch / arch / ISA propagation.
- **What** – full die state, every cycle.

## 3. CAPRI1 SoC & EMFI bench




| 12 756 FFs total | |
|---|---|
| Peripherals (UART, GPIO) | 2 722 |
| GPRs (x0 … x31) | 992 |
| µarch (pipeline, CSR) | 784 |
| SRAM tag + control | 8 258 |

- Pulse ±10 V, 4–10 ns – **just after rising edge**.
- CAPRI1: Ibex, 2×2 mm², 0.18 µm, 1.8 V.

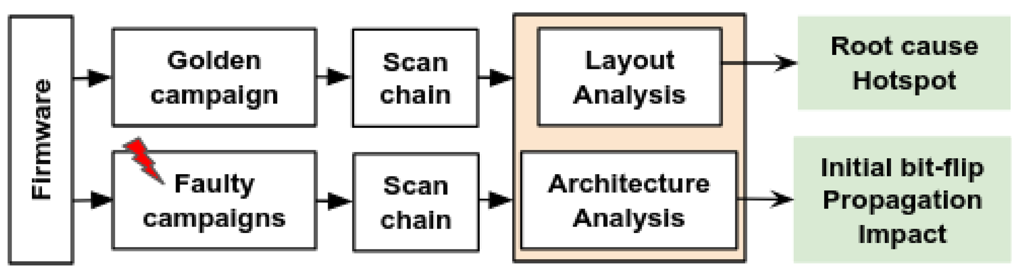
## Threat model

| | |
|---|---|
| **Attacker** | physical access; coil-on-die; one shot/cycle |
| **Granularity** | one specific clock cycle per campaign |
| **Goal** | flip a PIN-verification outcome |
| **Out of scope** | key probing, decapsulation |

## Why this matters

- **Cell-level pre-tape-out** fault budget.
- **PIN bypass** without key recovery.
- One framework – **physics to firmware**.

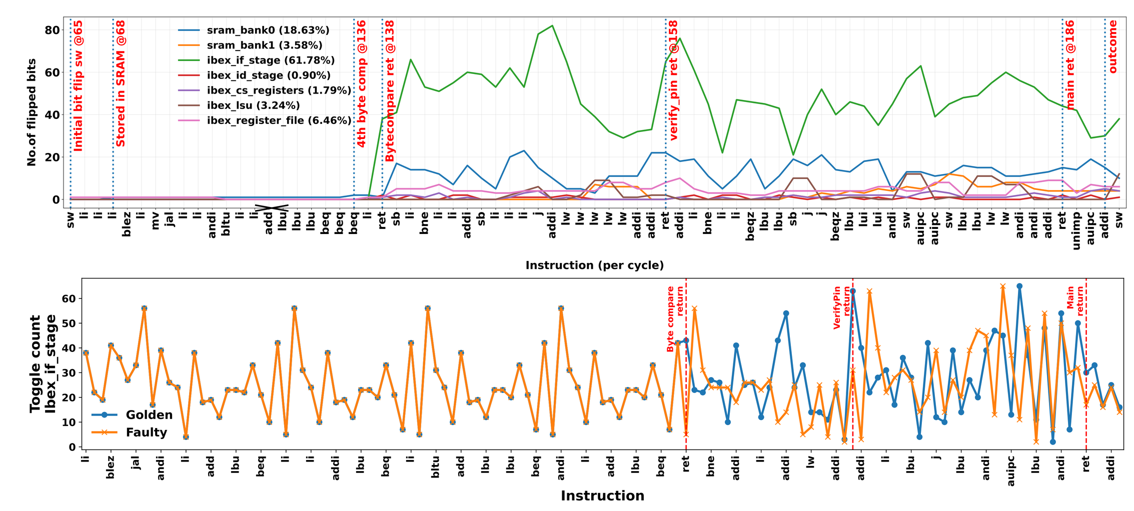
## 4. Hierarchical component-centric analysis




Each scan dump → **architecture** + **layout** layers → 4 steps → 1 label:

1. **Initial upset:** first deviating block.
2. **Propagation:** data → control flow.
3. **System:** crash / silent / bypass (*Detour, Tunnel, Spinner, Wormhole*).
4. **Label:** e.g. regfile-originated SDC w/ success.

## 5. Case study: VerifyPIN




**VP0** unprotected; **VP7** redundant compare.

- C.65 (VP0): 1-bit upset in `x15` at (302, 731) µm.
- Corrupted word re-written as *instruction* in `sram_bank0`; refetch → `0x04` into PIN-check reg.
- Misaligned PC ⇒ **apparent success on invalid PIN**.

**Fault propagation:**
~6 % regfile → ~19 % IMEM → ~**60 %** fetch.

## 6. Outcome & origin ($n$=331)

| Workload | Camp. | Reset+Crash | No-eff. | Succ. | Uncl. |
|---|---|---|---|---|---|
| VP0 (unprot.) | 191 | 70 | 118 | 2 | 1 |
| VP7 (hardened) | 140 | 47 | 91 | 0 | 2 |
| Total | 331 | 117 | 209 | 2 | 3 |

| First deviating component ($n$=65: crash 63 + success 2) | | | | | | | |
|---|---|---|---|---|---|---|---|
| Wkld | IF | DE | LSU | CSR | ALU | RegF. | SRAM |
| VP0 | 19 | 5 | 6 | 5 | 4 | 1 | 1 |
| VP7 | 10 | 4 | 4 | 3 | 2 | 1 | 0 |
| Total | 29 | 9 | 10 | 8 | 6 | 2 | 1 |

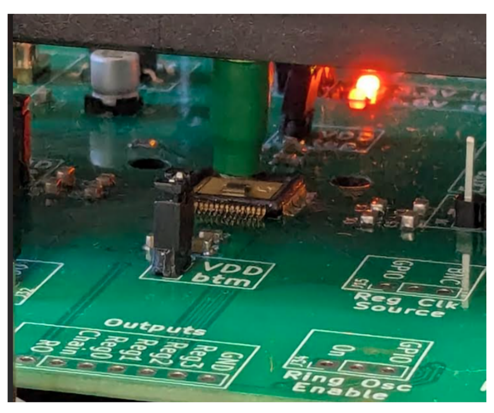
**VP7 redundant compare: 0/140 successes.**

**29/65 first flips strike the IF stage.**

## Scan tells us *where* – but not *why*

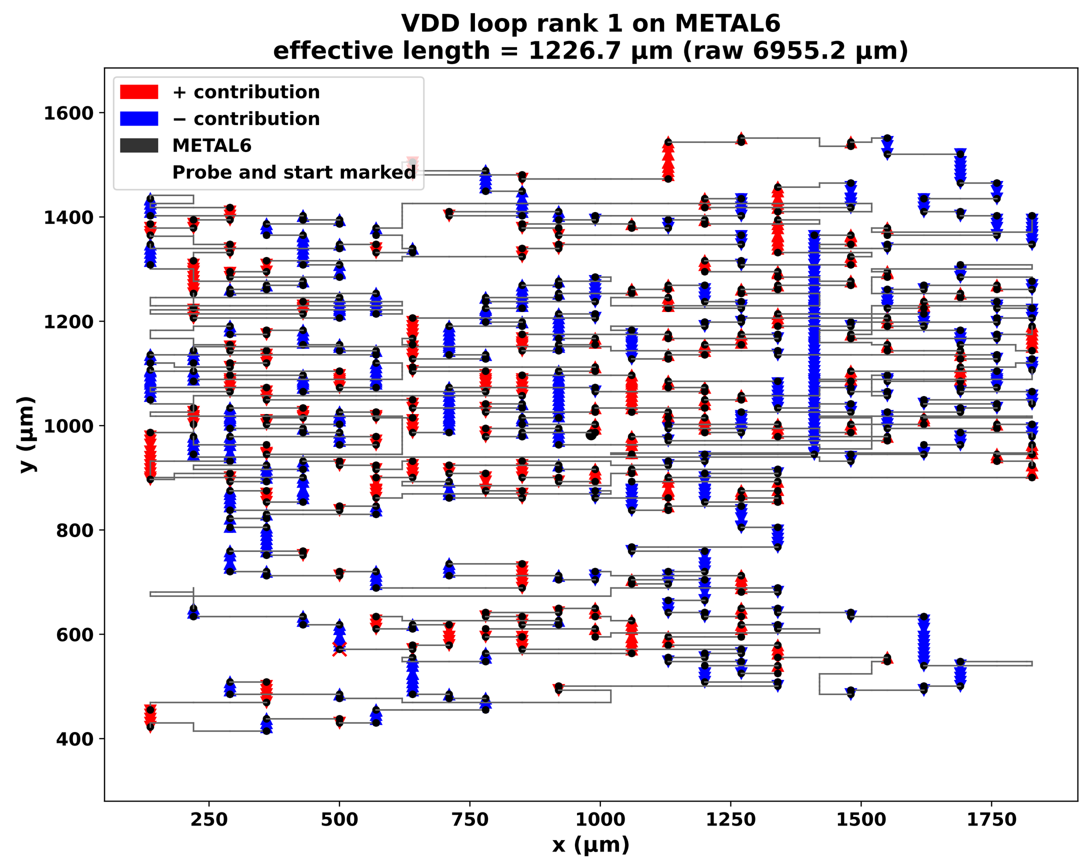
Scan pins the *cell*; to learn *why that FF, that cycle*, we model the **Power Delivery Network (PDN)** – the on-chip VDD/VSS mesh (M5/M6 grid + M3/M4 stripes). Layout-only, runs **before tape-out**.

## 7. How the bit flips: PDN loop coupling



- Coil → fast dΦ/dt over die.
- Eddy currents close *inside* the M5/M6 mesh.
- Each VDD/VSS rectangle = **one-turn loop**.
- Local $R$–$L$ ⇒ transient $\Delta V$ on the rails feeding nearby FFs.

## 8. Direction-aware cancellation



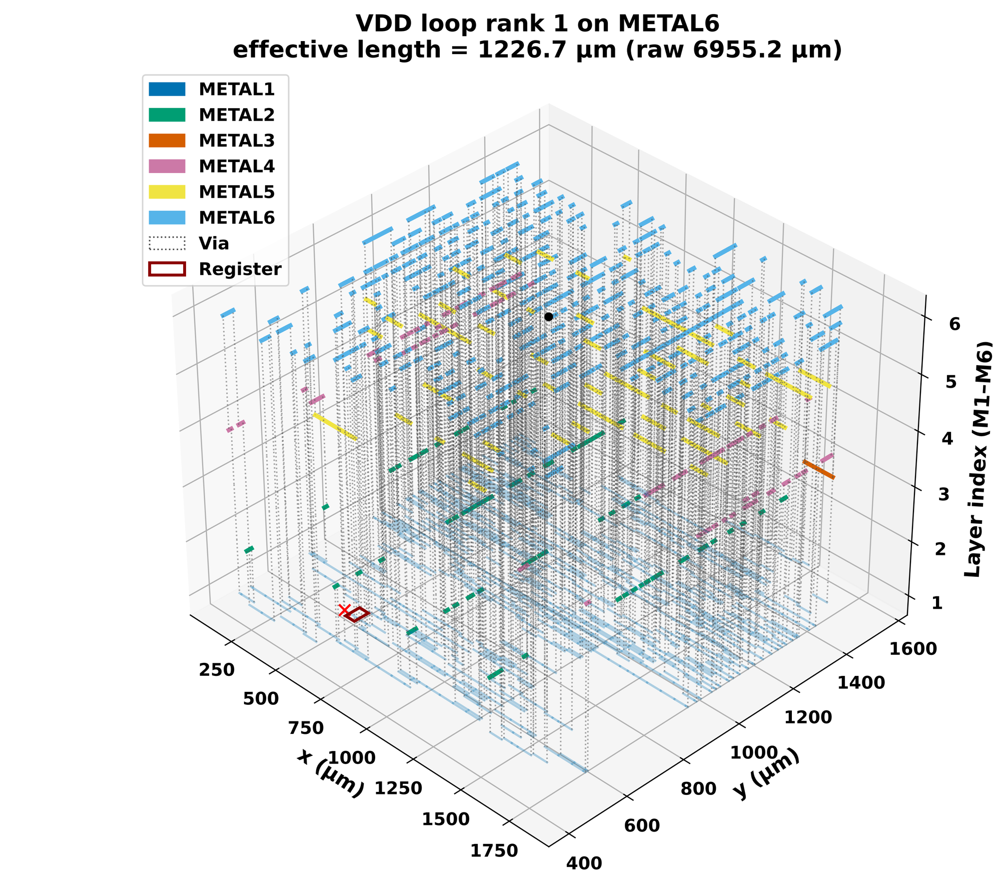

Interior loops *cancel*; only **boundary loops** contribute. Signed length:

$$L_{\text{eff}}(\ell;\mathbf{p}) = \left|\sum_{e\in\ell\cap \mathrm{M6}} \hat{\mathbf{t}}_e(\mathbf{p})\cdot(\mathbf{q}_1-\mathbf{q}_0)\right|$$

Top VDD loop, M6: 6 955 µm raw → $L_{\text{eff}}$=1 227 µm.

**82 % cancellation – one loop survives.**

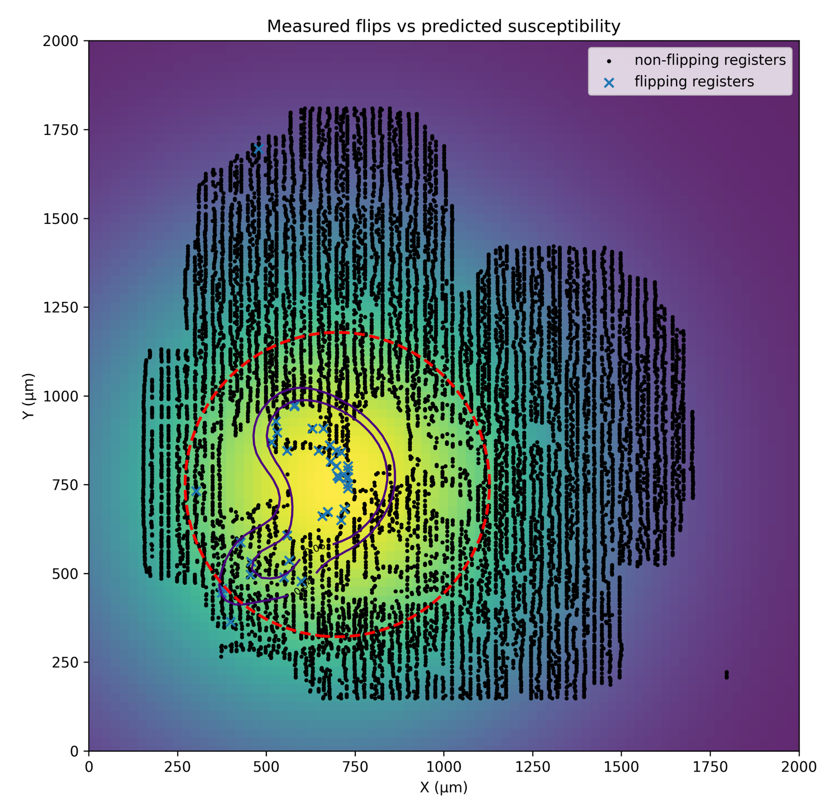
## 9. Loop → droop → bit-clear




Series $R$–$L$ bound (1 mm, 10 V / 4 ns): $|\Delta V_{\text{VDD}}|$=1.23 V, $|\Delta V_{\text{VSS}}|$=0.24 V ⇒ $|\Delta S|$=**1.47 V**. Margin ~**0.4 V**.

**1.47 V ≫ 0.4 V ⇒ bit clears.**

## 10. Simulation ↔ measurement




Heatmap (*pre-silicon*) + crosses (*post-silicon*).

**Top-20 %** ⇒ **29 / 41** (**70.6 %**). Hotspot = **`ibex_if_stage`** (= **29/65 in IF**).

## Take-aways

| | |
|---|---|
| **Where** | First flip = **IF stage**; $(x, y)$ by scan. |
| **How** | regfile → IMEM → fetch. |
| **Why** | Boundary VDD loop; $|\Delta S|$=**1.47 V**. |
| **When** | Pre-tape-out; top-20 % catches 70 %. |
| **Defence** | VP7 redundant compare: **0/140**. |


**Code, scan dumps, PDN scripts:** github.com/Secure-Embedded-Systems/capri1_ets • **Funding:** NSF CNS-2219810 • **Builds on:** CROC (Sauter *et al.*, arXiv 2025) & FaultDetective (Liu *et al.*, TCHES 2024).